Euler method

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$h = \frac{\text{Final value} - \text{Initial value}}{\text{Step}}$$

1) $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y=1$ for $x_0=0$ find y approximately for $x_0 = 0.1$ by Euler method.

Given $x_0 = 0$, $y_0 = 1$ $\left(y(\overset{x_0}{0}) = \overset{y_0}{1} \right)$

$y(0.1) = ?$

$$h = \frac{0.1 - 0}{5} \Rightarrow 0.02$$

Step 1 $\quad y_{n+1} = y_n + h f(x_n, y_n)$

$n=0$

$$y_1 = y_0 + 0.02\left(\frac{y_0 - x_0}{y_0 + x_0}\right)$$

$$= y_0 + 0.02\left(\frac{1-0}{1+0}\right)$$

$$= 1 + 0.02(1)$$

$\boxed{y_1 = 1.02}$ $\quad x_1 = x_0 + h$

$\boxed{x_1 = 0 + 0.02}$

Step 2 $n=1$

$$y_2 = y_1 + h f(x_1, y_1)$$

$$= 1.02 + 0.02\left(\frac{1.02 - 0.02}{1.02 + 0.02}\right)$$

$$\boxed{y_2 = 1.039} \qquad x_2 = x_1 + h$$

$$= 0.02 + 0.02$$

$$\boxed{x_2 = 0.04}$$

Step 3 $n=2$

$$y_3 = y_2 + h f(x_2, y_2)$$

$$= 1.039 + 0.02\left(\frac{1.039 - 0.04}{1.039 + 0.04}\right)$$

$$\boxed{y_3 = 1.058} \qquad x_3 = x_2 + h$$

$$= 0.04 + 0.02$$

$$\boxed{x_3 = 0.06}$$

Step 4

$$y_4 = y_3 + 0.02\left(\frac{1.058 - 0.06}{1.058 + 0.06}\right)$$

$$y_4 = 1.058 + 0.02 \left(\quad \right)$$

$$y_4 = 1.076 \qquad x_4 = 0.06 + 0.02 = 0.08$$

Step 5

$$y_5 = 1.076 + 0.02\left(\frac{1.076 - 0.08}{1.07 + 0.08}\right)$$

$$\boxed{y_5 = 1.093} \qquad \boxed{x_5 = 0.08 + 0.02 = 0.10}$$